Continuous-Time Chebyshev and Elliptic Filters

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Key Concepts

1) Chebyshev type I lowpass filter characteristics

    a) Equi-ripple magnitude response in the passband, monotone decreasing response in the stopband.

    b) Design parameters are the order N, ϵ, which determines the passband ripple, and Ω_o, which determines the passband edge frequency.

    c) The width of the transition from passband to stopband decreases as N increases.

2) Chebyshev type I lowpass filter design

    a) Set Ω_o equal to the passband edge frequency.

    b) Choose ϵ to satisfy the passband gain constraint.

    c) Choose N to satisfy the stopband constraint.

3) Chebyshev type II lowpass filter characteristics

    a) Monotone decreasing response in the passband, equi-ripple magnitude response in the stopband.

    b) Design parameters are the order N, ϵ, which determines the

stopband ripple, and Ω_o, which determines the stopband edge frequency.

c) The width of the transition from passband to stopband decreases as N increases.

4) Chebyshev type II lowpass filter design

a) Set Ω_o equal to the stopband edge frequency.

b) Choose ϵ to satisfy the stopband gain constraint.

c) Choose N to satisfy the passband gain constraint.

5) Elliptic lowpass filters

a) Equi-ripple magnitude response in the passband, equi-ripple magnitude response in the stopband.

b) The width of the transition from passband to stopband decreases as filter order N increases.

Video Screenshots

Butterworth - monotonic in pass/stop bands

Chebyshev Type I - ripple in pass band, monotonic in stop band

Chebyshev Type II - monotonic in passband, ripple in stop band

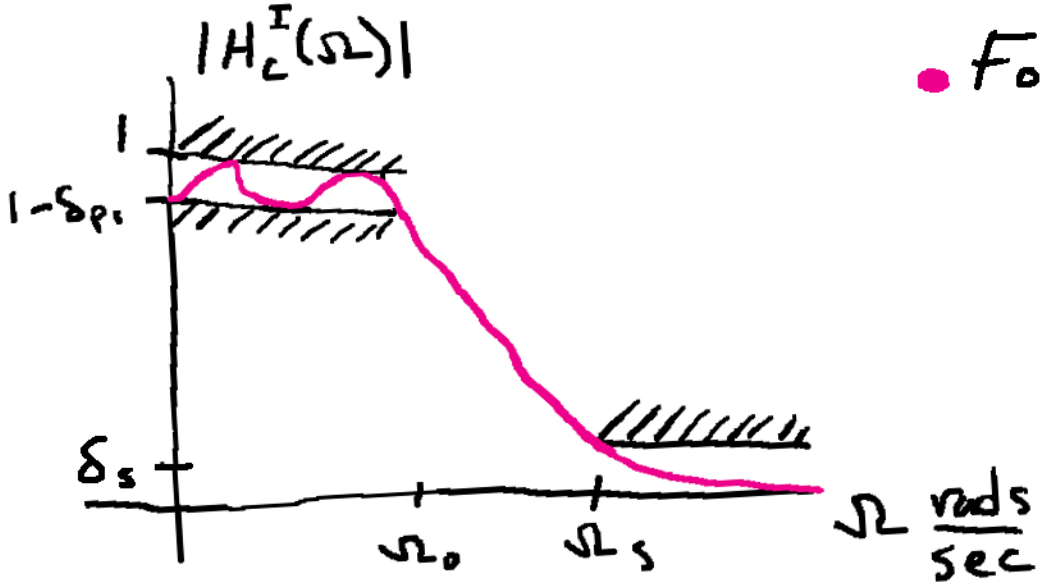
- Elliptic - ripple in both pass/stop bands

Chebyshev Type I

$$|H_c^I(\Omega)|^2 = \frac{1}{1+\varepsilon^2 T_N^2\left(\frac{\Omega}{\Omega_0}\right)}; \qquad T_N(x) = \begin{cases} x, & N=1 \\ 2x^2-1, & N=2 \\ \vdots \end{cases}$$

- For $\left|\frac{\Omega}{\Omega_0}\right| < 1$, $T_N\left(\frac{\Omega}{\Omega_0}\right)$ oscillates between ± 1, $T_N(1) = 1$

  $\Rightarrow |H_c^I(\Omega)|^2$ oscillates between $\frac{1}{1+\varepsilon^2}$ and 1



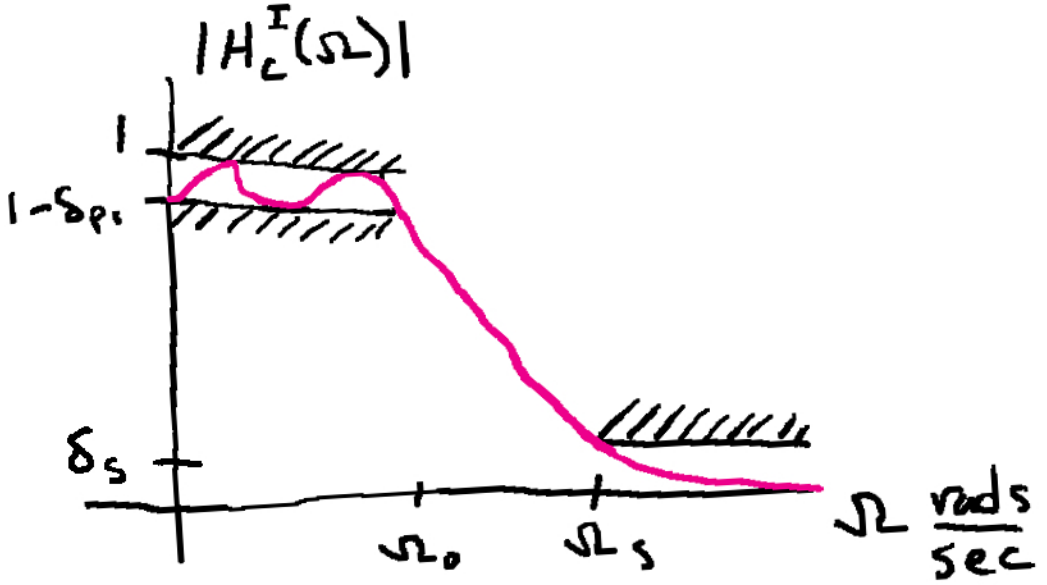
- For $\left|\frac{\Omega}{\Omega_0}\right| > 1$, $T_N\left(\frac{\Omega}{\Omega_0}\right)$ monotone increasing

$$|H_c^I(0)|^2 = \begin{cases} 1, & N \text{ odd} \\ \frac{1}{1+\varepsilon^2}, & N \text{ even} \end{cases}$$

Three design parameters: $N, \varepsilon^2, \Omega_0$

- Set Ω_0 = passband edge
- Choose ε^2 so $1 - \delta_{p_1} = \left(\frac{1}{1+\varepsilon^2}\right)^{1/2}$
- Choose N to satisfy stopband constraint

$$\delta_s^2 \geq \frac{1}{1 + \varepsilon^2 T_N^2\left(\frac{\Omega_s}{\Omega_0}\right)}$$



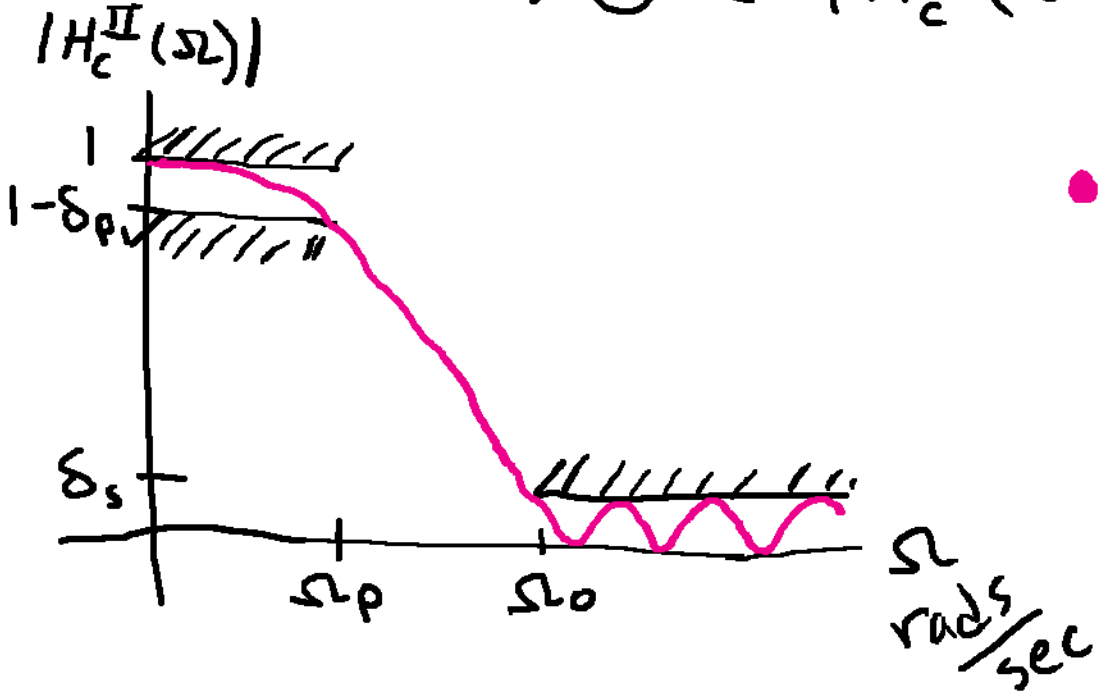
- Filter poles located on an ellipse in the s plane

Chebyshev Type II

$$|H_c^{II}(\Omega)|^2 = \frac{1}{1+\left[\varepsilon^2 T_N^2\left(\frac{\Omega_0}{\Omega}\right)\right]^{-1}} = \frac{\varepsilon^2 T_N^2\left(\frac{\Omega_0}{\Omega}\right)}{1+\varepsilon^2 T_N^2\left(\frac{\Omega_0}{\Omega}\right)}$$

- For $\left|\frac{\Omega_0}{\Omega}\right| < 1$ $(|\Omega| > \Omega_0)$, $T_N\left(\frac{\Omega_0}{\Omega}\right)$ oscillates between ± 1

$$\Rightarrow 0 \le |H_c^{II}(\Omega)|^2 \le \frac{\varepsilon^2}{\varepsilon^2+1}, \quad |\Omega| > \Omega_0$$



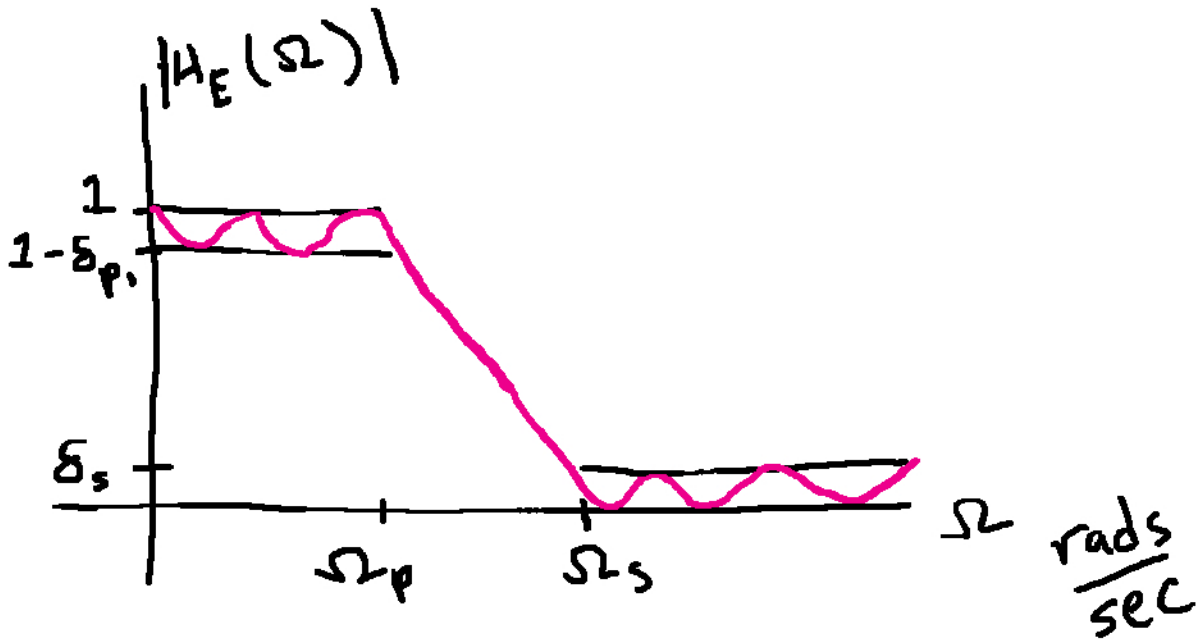
- For $|\Omega| < \Omega_0$, $T_N\left(\frac{\Omega_0}{\Omega}\right)$ increases as $\Omega \to 0$

$$|H_c^{II}(0)| = 1$$

Elliptic Filters

$$|H_E(\Omega)|^2 = \frac{1}{1+\varepsilon^2 U_N^2(\Omega)}$$; $U_N(\Omega)$ Jacobian elliptic function

equiripple error in both pass/stop bands



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